Emotion Detection System

Medusa Project

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Introduction

The **Medusa** project aims to create a desktop application capable of detecting human emotions in real-time using a webcam. Developed by a team from Athens University of Economics and Business, this system utilizes advanced machine learning to identify emotional states such as happiness, sadness, anger, and surprise. Medusa demonstrates the potential of emotion detection technology to enhance user experiences across various domains.

Market of Emotion Detection Systems

The **emotion detection market** is rapidly expanding as businesses and researchers recognize the value of real-time emotional awareness. These systems analyze **facial expressions, voice tones, and text messages** to detect emotions like happiness, sadness, anger, surprise, fear, disgust, contempt, and neutrality. **AI and deep learning** are enhancing emotion detection, expanding its use in **customer service, healthcare, marketing, and HR**.

Noldus FaceReader

Noldus FaceReader is a leading software for real-time facial expression analysis, widely used in academic research and consumer behavior testing.

MorphCast

MorphCast, an AI-powered Zoom app, tracks attention, engagement, and emotions in video calls, benefiting virtual meetings, e-learning, and remote work.

Hugging Face pretrained-models

Hugging Face pretrained models provide accurate emotion detection solutions, making it easier for businesses to integrate AI-driven emotional analytics. A notable Hugging Face project is **Facial Emotions Image Detection by dima806.**

Significance and Applications

Customer Service

Helps support teams assess customer satisfaction and adjust approaches for more personalized service.

Marketing & Sales

Analyzes customer emotions in real-time to tailor campaigns and products to user needs.

Education

Tracks student engagement and emotional well-being, assisting educators in providing personalized learning experiences.

Mental Health

Detects emotional distress for early intervention, supports therapy, and enables crisis assistance via apps and chatbots.

Targeted Marketing

Evaluates customer reactions to ads and interactions, optimizing content and recommendations.

Societal Impact

Emotionally intelligent systems, can create empathetic technology that enhances human-technology interactions.

Future Innovation

Integrating emotional intelligence into technology can make it more intuitive, supportive, and responsive across personal and professional domains.

Introduction

Emotion Detection System

01 An overview of the steps used to build and train our emotion detection model. — **Methodology**

02 Details on the dataset's composition and preprocessing for effective training. — **Dataset**

03 A look at the CNN architectures used and their role in detecting emotions. — **CNN Models**

04 Evaluation of model performance through metrics and sample predictions. — **Analysis**

Methodology

Methodology

Data Collection

Locate and select an appropriate dataset for emotion detection

Model Architecture

Identify suitable pre-trained models or build custom CNN architectures for image classification, tailoring them to detect facial expressions effectively.

Training and Evaluation

Train the models on the processed dataset and analyze performance metrics, focusing on accuracy and other relevant measures.

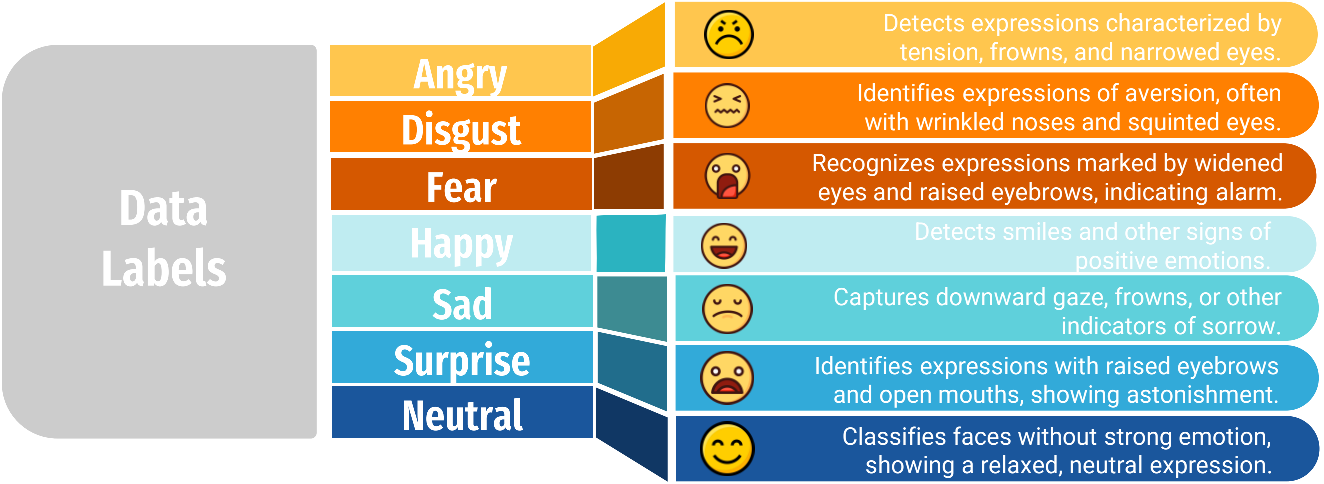
Documentation and Deployment

Document the project's workflow, model performance, and insights, and deploy the model in an application to demonstrate real-world functionality.

Dataset

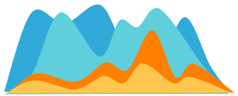
Dataset

The **FER-2013** dataset (35,887 images), designed specifically for emotion recognition tasks, consists of 48x48 pixel grayscale images of faces, making it suitable for compact and efficient model training. It includes seven emotion classes: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral, each representing distinct facial expressions to enhance model accuracy in emotion detection.

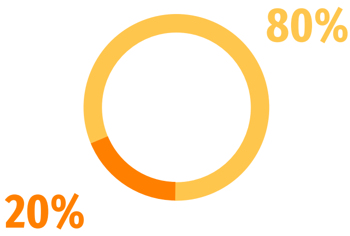


Dataset

The effective processing and augmentation of the dataset are crucial preparation steps for machine learning models, as the quality and variability in the data would play a significant role in the model's performance. In this regard, the FER-2013 dataset in the subsequent discussions has been subjected to various preprocessing and augmentation methods that could enhance the robustness and generalization capability of the emotion detection model.

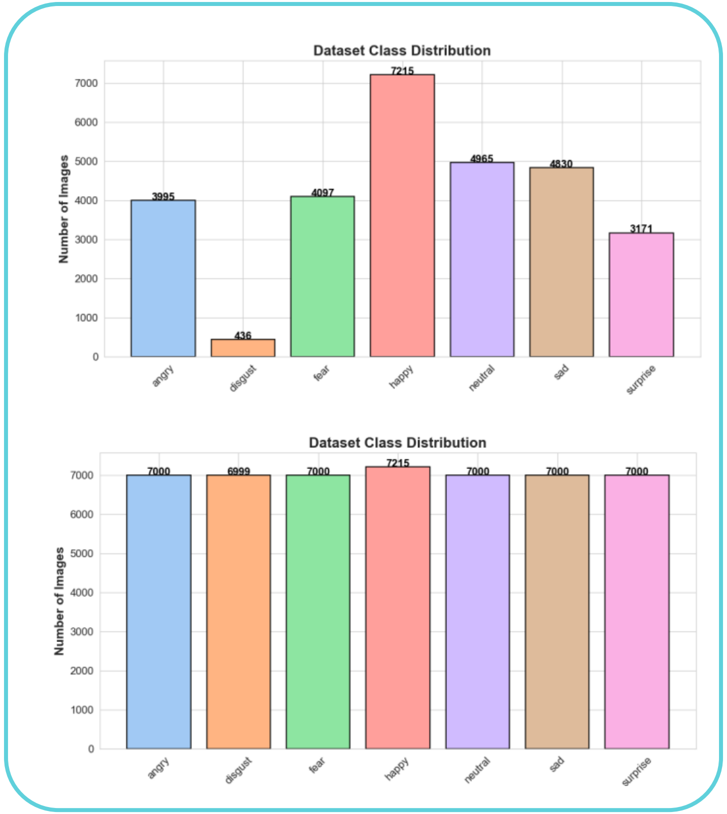


In data processing, we first normalized the image data to a 0-1 range to enhance model performance.



We split the dataset, allocating 80% for training and 20% for testing, ensuring a balanced evaluation of the model.

Dataset



Imbalanced Dataset

The dataset contains **7 classes** with varying image counts, as shown in the first plot.

Data Augmentation

To balance the dataset, we applied augmentation to generate images for underrepresented classes.

Target Distribution

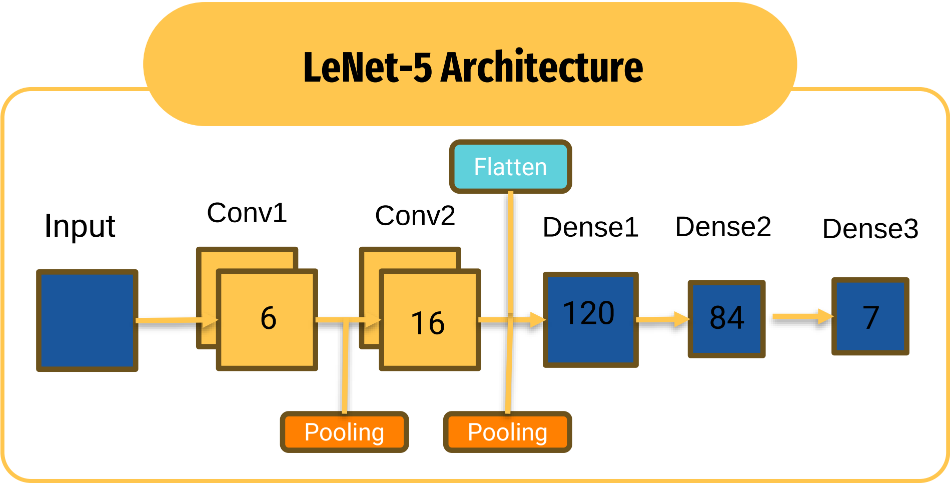
The goal was to have approximately **7,000 images per class** to ensure uniformity.

Augmentation Techniques

Used **ImageDataGenerator** with rotation (30°), shifts (10%), shear (0.2), zoom (20%), horizontal flip, and nearest fill mode.

CNN Models

CNN Models



Purpose

Simpler model for emotion detection, capable of identifying basic features with moderate accuracy

Two Convolutional Layers

Basic feature extraction layer that applies filters to capture edges and textures in facial images

Training & Validation Performance

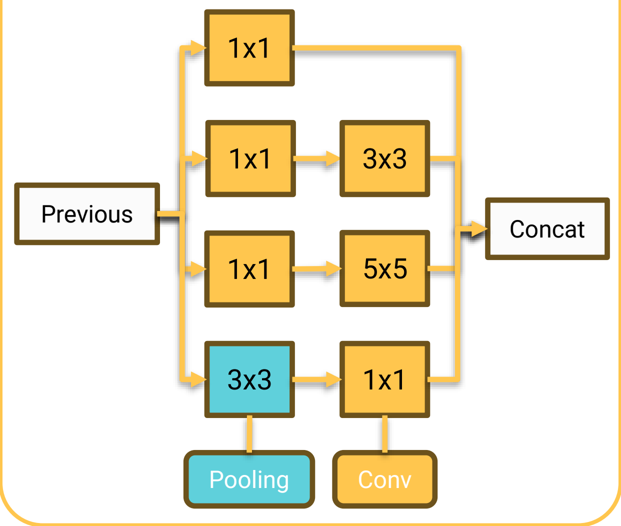
Modest accuracy (training: 51%, validation: 45%) but with good generalization

Efficiency and Stability

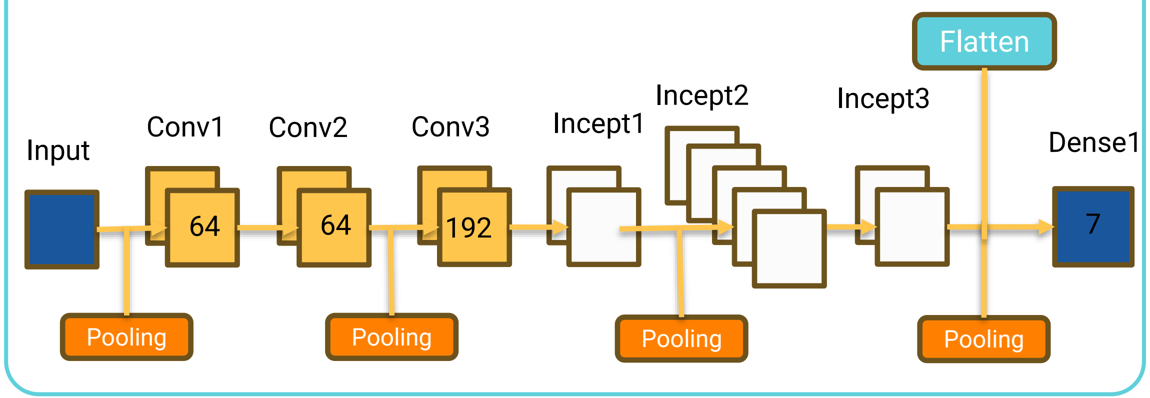
Despite lower accuracy , LeNet-5 offers a steady training process with limited overfitting

CNN Models

Inception



GoogleNet Architecture



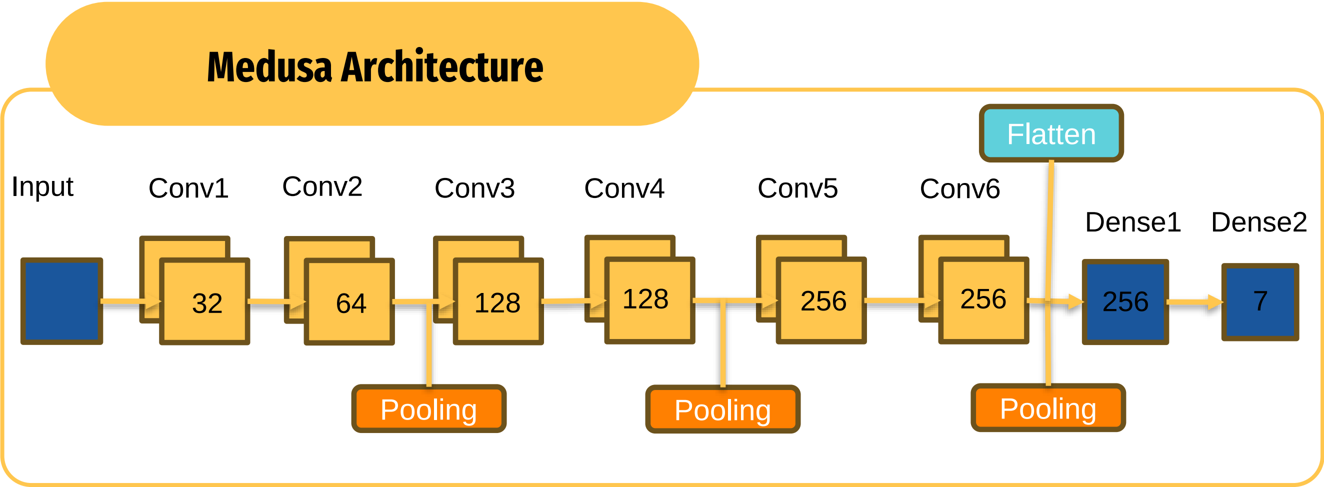
Deep Architecture with Inception Modules

GoogLeNet uses a deeper network with Inception modules, allowing it to capture complex features across various scales

Training & Validation Performance

Better training accuracy (61%) but encountered challenges with validation accuracy (46%), indicating tendency to overfit with complex patterns

CNN Models



Multi-Layer Convolutional Design

Six convolutional layers to progressively capture features, from basic edges to complex emotional expressions

Superior Accuracy

Medusa achieved higher training and validation accuracy

Dropout for Regularization

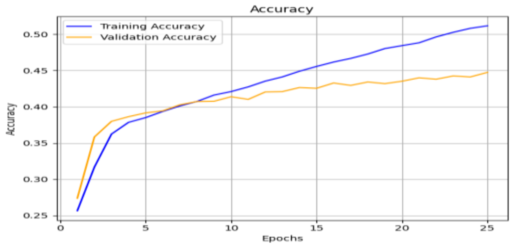
Dropout layers help prevent overfitting

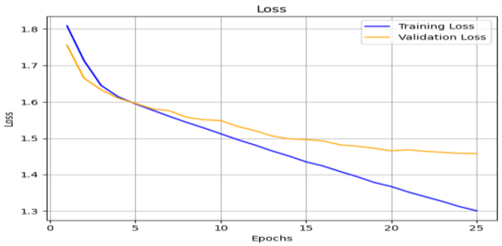
Real-Time Performance

Designed for real-time processing, making it suitable for desktop applications

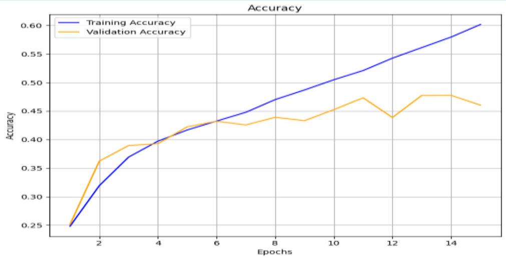
Quantitative and Qualitative Analysis

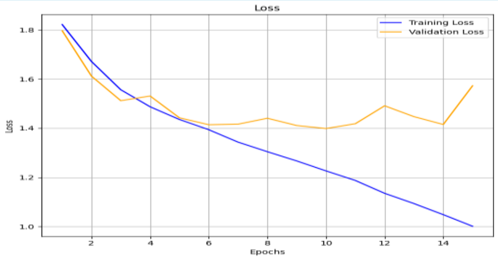
Quantitative Analysis



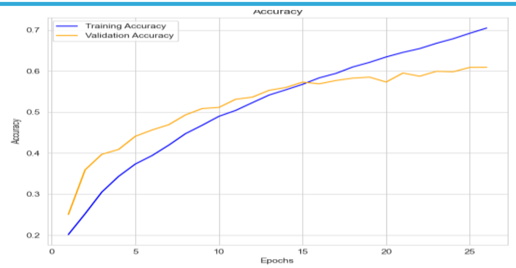


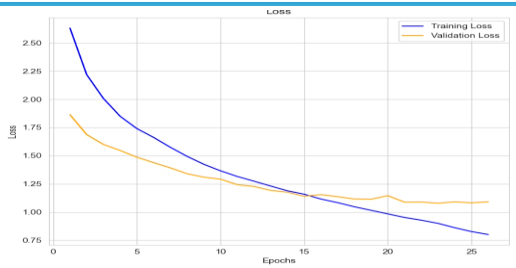
LeNet-5





GoogleNet





Medusa

Quantitative Analysis

	Training accuracy	Training loss	Validation accuracy	Validation loss
LeNet-5	0.51	1.30	0.45	1.45
GoogLeNet	0.61	1.00	0.46	1.59
Medusa	0.72	0.75	0.6	1.13

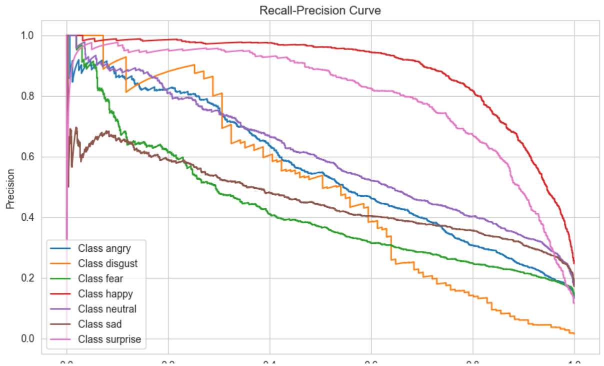
Medusa

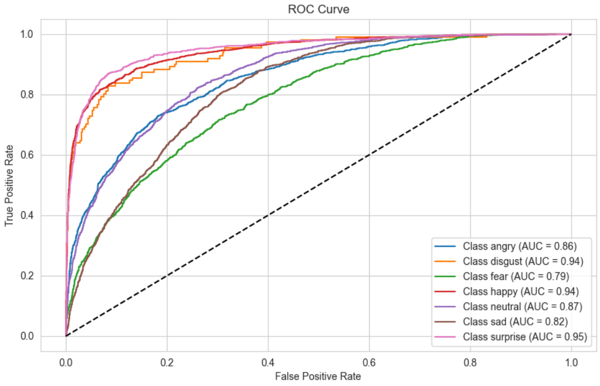
Demonstrated the best performance, with a training accuracy of 72% and a validation accuracy of 60%, showing a stronger generalization to unseen data.

GoogLeNet

Performed better than LeNet-5, reaching 61% training accuracy and 46% validation accuracy. However, the gap between training and validation performance suggests possible overfitting.

Quantitative Analysis





Precision-Recall Curve

The "**happy" (pink) and "surprise" (purple) classes** show the **highest precision across recall values**, indicating that these emotions are predicted with **high confidence and fewer false positives**.

The "**fear" (green) and "neutral" (brown) classes** have the **worst PR curves**, meaning they suffer from **lower precision** as recall increases, suggesting the model struggles to correctly identify these emotions without introducing many false positives.

ROC Curve

The "**surprise" (purple) and "disgust" (orange) classes** have the **highest AUC values (0.95 and 0.94, respectively)**, showing that the model is highly effective at distinguishing these emotions from the others.

The "**fear" (green) class has the lowest AUC (0.79)**, meaning the model struggles to separate fear from other emotions, likely due to overlapping features with similar emotions like sadness or anger.

Quantitative Analysis

Overall Performance

The model achieves an **accuracy of 59.19%**, with an **F1-score of 0.5849**, indicating moderF1-scoreate performance in classifying emotions. The macro (0.55) is lower than the weighted F1-score (0.58), showing that some classes perform significantly worse than others.

Best Performing Class

The "happy" class has the highest scores (**precision: 0.82, recall: 0.79, F1-score: 0.81**), meaning the model is very reliable in identifying happy expressions.

Efficiency and Stability

"Fear" has the lowest **F1-score (0.33)** due to very low **recall (0.24),** suggesting that many fearful expressions are misclassified as other emotions.

Best Evaluation Approaches

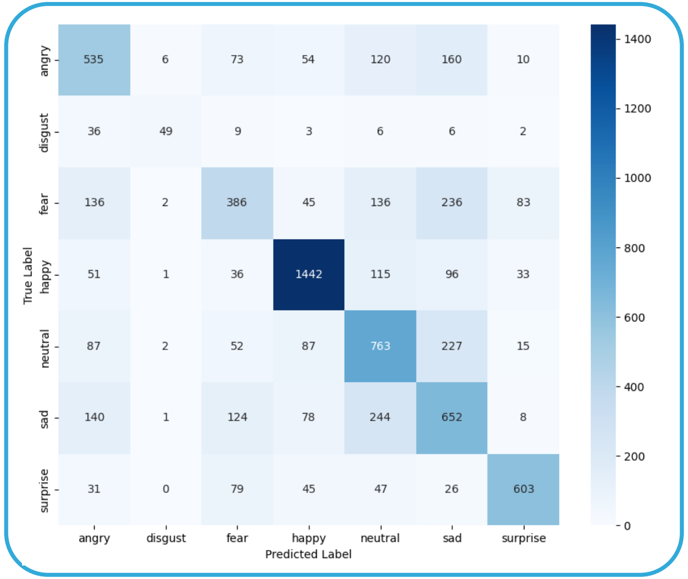
We should use balanced metrics instead of accuracy since it is misleading in imbalanced datasets. Instead, we should take into account the **F1-score, Precision and Recall**.

Classification Report

	precision	recall	f1-score	support
angry	0.50	0.52	0.51	958
disgust	0.38	0.59	0.46	111
fear	0.53	0.24	0.33	1024
happy	0.82	0.79	0.81	1774
neutral	0.50	0.64	0.56	1233
sad	0.45	0.49	0.47	1247
surprise	0.72	0.75	0.74	831
accuracy			0.59	7178
macro avg	0.56	0.58	0.55	7178
weighted avg	0.60	0.59	0.58	7178

F1-score: 0.5849
Precision: 0.5971
Recall: 0.5919
Accuracy: 0.5919

Qualitative Analysis



Strong Performance

The Medusa model excelled in recognizing 'happy' emotions, with 1.442 correct classifications

Challenges

Difficulties in distinguishing 'fear' from 'angry', 'neutral', and 'sad' emotions were observed

Areas for Improvement

The model struggled with 'disgust' detection, often misclassifying it as 'angry' or 'fear'

Tech Stack

Deep Learning Framework: Tensorflow, Keras

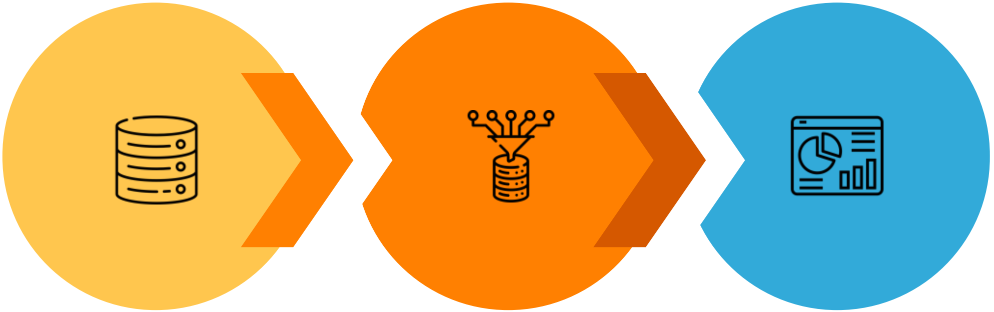
Data Handling: Numpy, Scikit Learn

Computer Vision: OpenCV

GUI Framework: Tkinter

Future Work and Improvements

Enhanced Model | **Feature Expansion** | **End-to-End System**



Enhanced Model

Improve accuracy and generalization through advanced regularization techniques and ensemble learning

Feature Expansion

Integrate additional models for detecting age, gender and specific facial expressions

End-to-End System

Develop Medusa into a comprehensive solution for emotion analytics and intelligent human-computer interaction

Thank you !